



Signature-based detection with YARA

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1. Objective

This report describes the creation, execution, and evaluation of a YARA rule aimed at detecting the EICAR test files, including their compressed versions. The EICAR file (eicar.com) is a standard test file used to verify the proper functionality of antivirus software. The rule targets both the original file and its compressed formats (eicar.zip and eicar2.zip), which are available in the lab directory.

The YARA rule was designed, implemented, and tested on the Lab VM, ensuring accurate detection without false positives or negatives.

2. YARA Rule Development

The YARA rule used to detect the EICAR files is as follows:

rule Code_EICAR

```
{
  meta:
    description = "Detect EICAR test files: eicar.com, eicar.zip, eicar2.zip"
    author = "Ahmed Mahfooz Ali Khan"
    date = "2024-10-24"

  strings:
    $eicar_com = "X5O!P%@AP[4\\PZX54(P^7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*"
    $eicar_zip_header = { 50 4B 03 04 } // ZIP file header (PK..)

  condition:
    ( $eicar_com and filesize < 200KB ) or
    ( $eicar_zip_header in (0..filesize) and filesize < 500KB )
}
```

Rule Breakdown:

- **Meta Section:**
 - **description:** Identifies the purpose of the rule (detect EICAR test files).
 - **author:** Specifies the rule's author (Ahmed Mahfooz Ali Khan).
 - **date:** Indicates when the rule was created (2024-10-24).
- **Strings Section:**
 - **\$eicar_com:** This string matches the known content of the EICAR test file.
 - **\$eicar_zip_header:** This is a hexadecimal signature representing the ZIP file header (50 4B 03 04), commonly seen in ZIP archives. It helps identify compressed EICAR files.
- **Condition Section:**
 - The condition first checks for the presence of the \$eicar_com string in files smaller than 200KB (to ensure efficiency and avoid unnecessary checks on large files).
 - For ZIP files, it looks for the ZIP header anywhere in the file (in (0..filesize)) as long as the file size is under 500KB.

This rule ensures that both the uncompressed and compressed EICAR test files are detected without flagging other files.

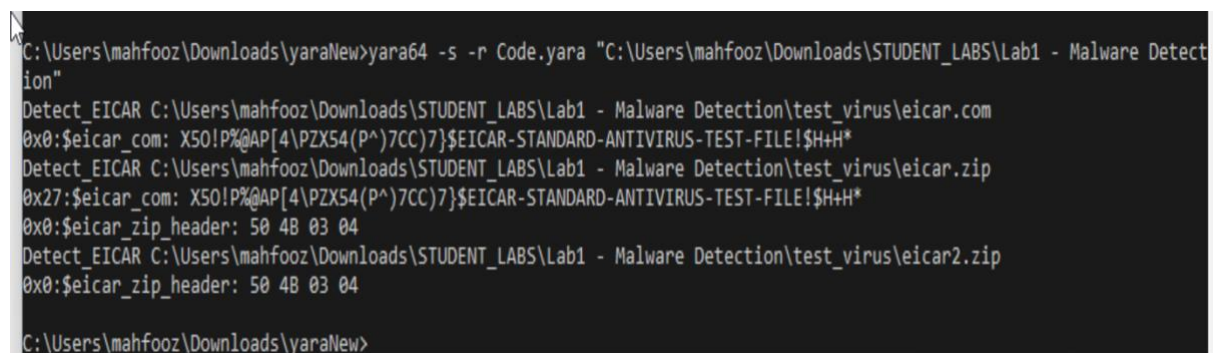
3. Execution Process

Command Used:

The following command was used to run the YARA rule on the Lab VM:

```
yara64 -s -r Code.yara "C:\Users\mahfooz\Downloads\STUDENT_LABS\Lab1 - Malware Detection"
```

Output of the Command:



```
C:\Users\mahfooz\Downloads\yaraNew>yara64 -s -r Code.yara "C:\Users\mahfooz\Downloads\STUDENT_LABS\Lab1 - Malware Detection"
Detect_EICAR C:\Users\mahfooz\Downloads\STUDENT_LABS\Lab1 - Malware Detection\test_virus\eicar.com
0x0:$eicar_com: X50!P%AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
Detect_EICAR C:\Users\mahfooz\Downloads\STUDENT_LABS\Lab1 - Malware Detection\test_virus\eicar.zip
0x27:$eicar_com: X50!P%AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
0x0:$eicar_zip_header: 50 4B 03 04
Detect_EICAR C:\Users\mahfooz\Downloads\STUDENT_LABS\Lab1 - Malware Detection\test_virus\eicar2.zip
0x0:$eicar_zip_header: 50 4B 03 04
C:\Users\mahfooz\Downloads\yaraNew>
```

Explanation of the Output:

- **eicar.com:**
 - Detected successfully at memory offset 0x0 based on the \$eicar_com string.
- **eicar.zip:**
 - Detected at memory offset 0x27 based on the \$eicar_com string.
 - Detected at offset 0x0 with the ZIP header \$eicar_zip_header (50 4B 03 04).
- **eicar2.zip:**
 - Detected at offset 0x0 using the ZIP header \$eicar_zip_header.

The YARA rule successfully matched the eicar.com file and its compressed versions based on both string and ZIP header patterns.

4. Test Results

False Negative Test:

The following files were tested:

- eicar.com
- eicar.zip
- eicar2.zip

Results:

- The YARA rule successfully detected all three files:
 - eicar.com
 - eicar.zip
 - eicar2.zip

There were **no false negatives**.

False Positive Test:

The rule was also tested on other files in the directory C:\STUDENT_LABS\Lab1 - Malware Detection to ensure no false positives occurred.

Results:

- The YARA rule did not detect any non-EICAR files, confirming that there were **no false positives**.

Observations

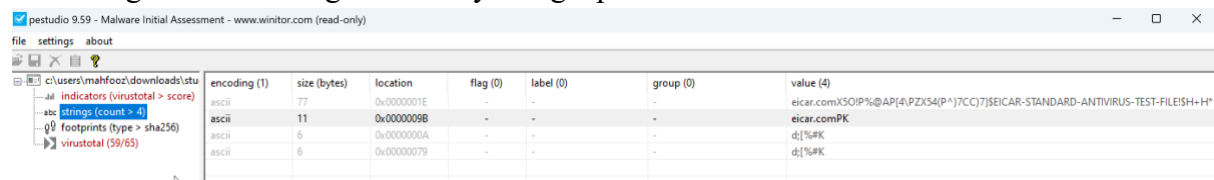
- The rule correctly identified the EICAR test files in both uncompressed (eicar.com) and compressed formats (eicar.zip and eicar2.zip).
- No unrelated files were mistakenly flagged, which confirms that the rule does not generate false positives.
- The detection string \$eicar_string works efficiently for both the uncompressed and archived formats.

5. Conclusion

The YARA rule was tested successfully and met all objectives. It accurately detected the EICAR test file and its compressed versions (eicar.zip and eicar2.zip) without generating false negatives or false positives. This demonstrates the reliability of the rule for detecting test malware files in the lab environment.

6. Reference

The string that has been generated by using a pestudio tool



encoding (1)	size (bytes)	location	flag (0)	label (0)	group (0)	value (4)
ascii	77	0x0000001E	-	-	-	eicar.comXSO!P%AP(4,PZ(X4(P*)7CC)7)EICAR-STANDARD-ANTIVIRUS-TEST-FILE!SH+H*
ascii	11	0x0000009B	-	-	-	eicar.comPK
ascii	6	0x0000000A	-	-	-	d;!%#K
ascii	6	0x00000079	-	-	-	d;!%#K

Report Completed by:

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